

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA1017	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	Mukesh Raj L	Reg. No.:	192571036
Date:		Duration:	60 minutes

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

AI-BASED SMART PHARMACEUTICAL MANUFACTURING

QUALITY ASSURANCE PLATFORM

Research Paper Review / Literature Survey Report

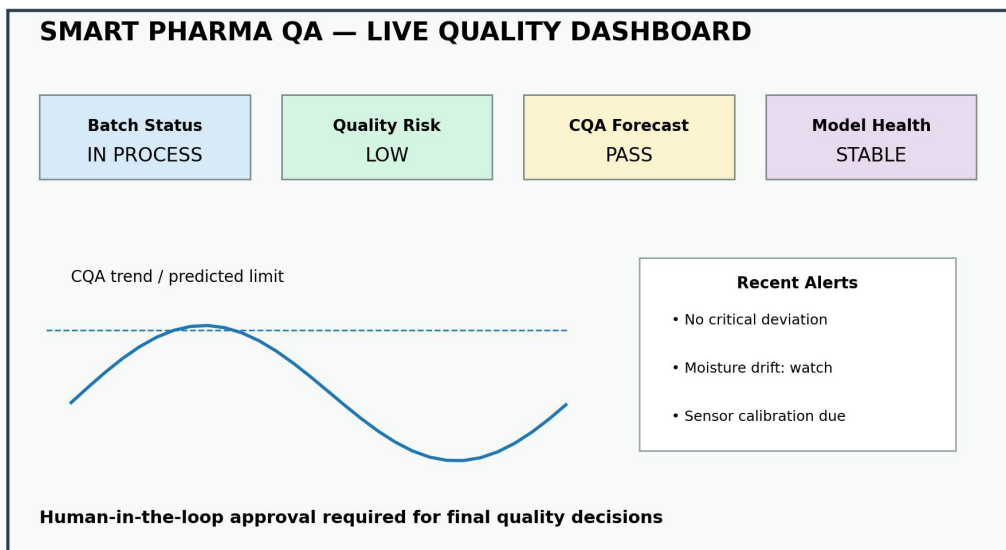


Figure 1. Conceptual smart quality assurance dashboard for AI-assisted pharmaceutical manufacturing.

1. ABSTRACT

Pharmaceutical manufacturing requires strict control of product quality, process variability, equipment performance, documentation, and regulatory compliance. Conventional quality assurance is often dependent on periodic sampling, laboratory testing, manual review, and retrospective deviation investigation. These approaches may detect quality problems only after process drift has already occurred. This literature survey examines how Artificial Intelligence (AI), Machine Learning (ML), Process Analytical Technology (PAT), Quality by Design (QbD), Internet of Things (IoT), computer vision, digital twins, and explainable AI can support a proactive quality assurance platform for pharmaceutical manufacturing. The reviewed literature indicates that AI can be used for critical quality attribute prediction, multivariate process monitoring, anomaly detection, image-based inspection, predictive maintenance, and adaptive decision support. However, significant barriers remain, including limited high-quality datasets, data silos, poor model interpretability, uncertainty, validation requirements, cybersecurity, data integrity, and difficulties in transferring models across products and manufacturing scales. Based on the literature, this report proposes an AI-based smart platform that integrates manufacturing data, AI/ML models, explainability, risk prioritization, human oversight, and audit-ready governance. The major research gap identified is the lack of a unified, validated, end-to-end framework that combines prediction, explanation, uncertainty, and compliant lifecycle management for practical pharmaceutical quality assurance.

Keywords

Artificial Intelligence, Machine Learning, Pharmaceutical Manufacturing, Quality Assurance, Quality by Design, Process Analytical Technology, Digital Twin, Predictive Quality, Explainable AI, Industry 4.0, GMP.

2. INTRODUCTION

The pharmaceutical sector operates under stringent quality requirements because manufacturing deviations can directly affect product safety, efficacy, and patient outcomes. Quality assurance therefore extends beyond final product testing and includes systematic control of raw materials, process parameters, equipment, environmental conditions, in-process measurements, documentation, deviations, and release decisions. Modern manufacturing increasingly generates large volumes of heterogeneous data through PAT instruments, sensors, laboratory information systems, manufacturing execution systems, and visual inspection systems.

The emerging Industry 4.0 environment creates an opportunity to transform this data into actionable quality intelligence. Instead of waiting for an out-of-specification result, an AI system can learn relationships between critical process parameters (CPPs) and critical quality attributes (CQAs), estimate future quality risk, detect abnormal patterns, and provide early warnings. Recent reviews emphasize predictive process modeling, real-time PAT, hybrid mechanistic–ML models, digital twins, and AI-enabled quality control as important directions for pharmaceutical manufacturing.

AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform

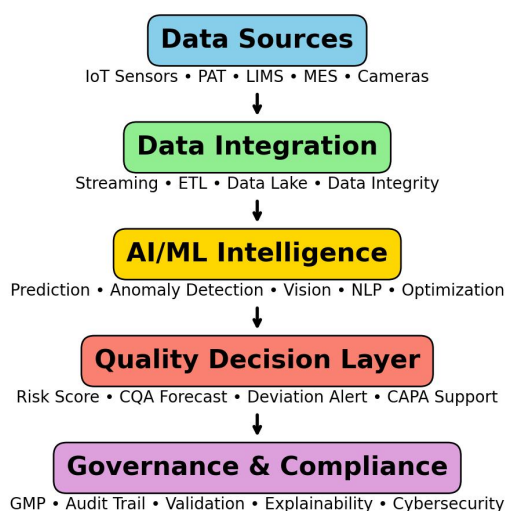


Figure 2. Proposed layered architecture of the AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform.

3. PROBLEM STATEMENT

Traditional pharmaceutical quality assurance processes can be fragmented and reactive. Data may be distributed across independent systems, while deviation analysis may require manual review of batch records and historical information. This can delay root-cause investigation and limit the ability to predict quality failures before they occur. Therefore, the problem addressed in this survey is how to design an AI-based platform that integrates real-time manufacturing information and historical quality data to predict quality outcomes, detect deviations, support risk-based decisions, and maintain traceable, explainable, and human-supervised quality assurance.

4. OBJECTIVES OF THE LITERATURE SURVEY

1. To review recent and relevant research on AI, ML, PAT, predictive quality, digitalization, and digital twins in pharmaceutical manufacturing.
2. To summarize the objectives, methodologies, technologies, datasets, and findings of selected studies.

3. To compare existing approaches using parameters such as predictive capability, real-time applicability, interpretability, scalability, and governance relevance.
4. To identify limitations and unresolved challenges in current AI-based pharmaceutical quality systems.
5. To identify a clear research gap for an integrated smart quality assurance platform.
6. To propose a practical system architecture and future research directions.

5. BACKGROUND CONCEPTS

5.1 Quality by Design (QbD)

QbD promotes systematic development based on scientific understanding and risk management. The approach identifies quality targets and investigates how material attributes and process parameters influence the final product. For an AI platform, QbD provides the domain structure for selecting meaningful CPPs, CQAs, and risk variables rather than treating manufacturing data as an unstructured collection of signals.

5.2 Process Analytical Technology (PAT)

PAT supports timely measurement and control of critical material and process attributes. Spectroscopy, moisture sensors, particle analysis, and other process measurements can provide high-frequency information that is suitable for multivariate analytics and machine learning. AI can complement PAT by converting complex sensor patterns into predicted quality indicators or early deviation warnings.

5.3 Critical Process Parameters and Critical Quality Attributes

CPPs are process variables that should be controlled because their variation can affect quality. Examples may include temperature, mixing speed, compression force, feed rate, moisture, and drying time. CQAs are measurable product properties that should remain within acceptable limits, such as assay, dissolution, content uniformity, impurity level, tablet weight, and hardness. A major AI task is learning the relationship $\text{CPP} \rightarrow \text{process state} \rightarrow \text{CQA}$.

5.4 Explainable and Human-Centred AI

Pharmaceutical decisions require traceability and scientific justification. Explainable AI techniques can identify influential variables and help quality professionals understand why a model produced a particular prediction. The proposed platform therefore treats AI as decision support rather than an unrestricted replacement for expert approval.

6. LITERATURE SURVEY AND INDIVIDUAL PAPER REVIEWS

6.1 Tercan and Meisen (2022): Predictive Quality in Manufacturing

This systematic review investigated machine learning and deep learning methods for predictive quality across manufacturing. Its relevance to pharmaceutical production lies in the general framework for linking process data with quality outcomes. The study highlights the potential of ML and DL to improve predictive quality assurance but also notes practical issues related to data availability, heterogeneous data, model deployment, and the difficulty of translating research models into production environments. The paper supports the use of predictive analytics as a foundation for proactive quality management, while also showing that model performance alone is insufficient for operational adoption.

6.2 Application of Artificial Neural Networks in PAT of Pharmaceutical Manufacturing (2022)

This review focuses specifically on artificial neural networks within the PAT concept for pharmaceutical manufacturing. It examines how neural networks can model complex nonlinear relationships in pharmaceutical unit operations and how sensor-based information can support monitoring and control. The paper is important because pharmaceutical processes often exhibit nonlinear and multivariable behavior that may not be represented well by simple linear models. The limitations identified include real-time implementation complexity, limited interpretability, data requirements, and the need for reliable integration with industrial process systems.

6.3 Emerging Digital Innovations in Pharmaceutical Manufacturing Quality (2025)

This systematised review evaluated digital innovations relevant to pharmaceutical manufacturing quality and identified themes including PAT and real-time release testing, AI, process modeling, IoT, and data management. The review reports opportunities for predicting product quality and variability, deviation root-cause analysis, real-time monitoring, and adaptive control. Important challenges include limited technical expertise, data security risks, validation, and potential inaccuracy. This study strongly supports the need for an integrated platform because the technologies are often discussed as separate components rather than as a single governed quality architecture.

6.4 Current State of Machine Learning Implementation in Pharmaceutical Process Modeling for Oral Solid Dosage Forms (2026)

This recent review describes ML applications across pharmaceutical unit operations and PAT for oral solid dosage manufacturing. It emphasizes prediction of quality-related variables, process optimization, and adaptive real-time frameworks. The review also identifies hybrid ML–mechanistic models and digital twins as important future directions. Major barriers include limited datasets, interpretability, extrapolation beyond training conditions, uncertainty quantification, transferability, and integration with existing workflows. These limitations directly motivate a platform with model health monitoring and human review.

6.5 Artificial Intelligence in Pharmaceutical Manufacturing: Applications, Case Studies, and GxP Implementation Considerations (2026)

This work discusses AI and ML applications for process control, real-time monitoring, inspection, soft sensors, multivariate analysis, anomaly detection, advanced control, and computer vision. Its key contribution is connecting technical AI capability with GxP implementation considerations. The study reinforces that successful deployment requires both predictive performance and compliance-oriented understanding of GMP requirements. This is highly relevant to the proposed system because an AI model should be embedded in a validated governance process rather than deployed as an isolated algorithm.

6.6 Digital and AI-Enabled Models in Pharmaceutical Development and Manufacturing (2026)

This regulatory-focused industry survey examines the increasing use of digital and AI-enabled models across pharmaceutical development and manufacturing. The work is valuable for understanding adoption barriers because industrial deployment depends on organizational, technical, validation, and regulatory factors. It supports the inclusion of auditability, lifecycle management, and regulatory readiness as platform design requirements rather than afterthoughts.

6.7 PharmaQAI: Machine Learning for Tablet Compression Quality Prediction (2026)

This recent preprint presents a machine learning framework for pharmaceutical tablet compression using 1,005 batches, 27 process parameters, and six quality attributes. Multiple regression and classification algorithms were evaluated with randomized hyperparameter optimization and five-fold cross-validation.

The work demonstrates the practical value of process-data-driven quality prediction and interactive visualization. However, as a preprint and product-specific implementation, its findings should be interpreted as an emerging case study rather than a universal validated solution. Transferability to other products, sites, and processes remains an open challenge.

Smart Quality Assurance Across the Manufacturing Line

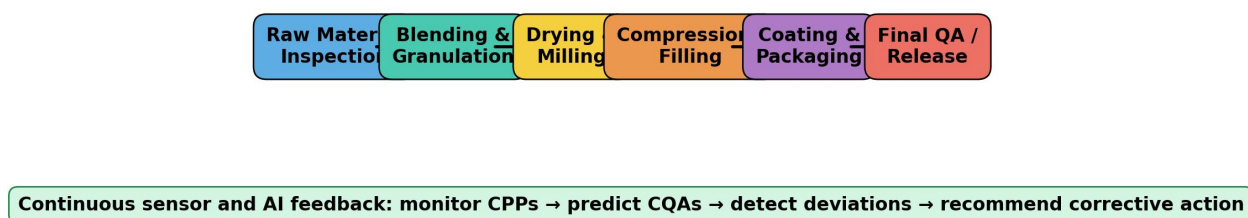


Figure 3. Manufacturing stages where AI-assisted quality monitoring can provide continuous feedback.

7. COMPARATIVE LITERATURE SURVEY TABLE

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Tercan & Meisen, 2022	Predictive quality	Systematic ML/DL review	Manufacturing quality studies	Predictive QA potential	Deployment and data heterogeneity	Production-ready governed frameworks
Markl et al., 2022	ANNs in pharma PAT	Review of ANN applications	PAT sensors; published studies	Models nonlinear relations	Interpretability; real-time integration	XAI and validated workflows
Digital Innovations, 2025	Digital quality technologies	Systematised review	38 studies; scholarly databases	AI/PAT/IoT enable proactive quality	Skills, security, validation	Integrated architecture
Rezaeizadeh et al.,	ML process	Review of unit	Oral solid dosage studies	Prediction and	Data, uncertainty,	Hybrid models

2026	modeling	operations/PAT		adaptive modeling	transferability	
AI in Pharma, 2026	AI and GxP	Review and case studies	Manufacturing/inspection use cases	Soft sensors, anomaly detection, vision	Compliance burden	GMP-aligned lifecycle
Cook et al., 2026	AI model adoption	Regulatory-focused industry survey	Industry survey	Digital adoption across CMC	Governance barriers	Regulatory-ready model management
PharmaQA I, 2026	Tablet quality prediction	Supervised ML platform	1,005 batches; 27 parameters; 6 CQAs	Practical predictive modeling	Product-specific preprint	Cross-site validation

8. COMPARATIVE ANALYSIS

The selected studies demonstrate a clear progression from isolated predictive models toward integrated digital manufacturing ecosystems. Early and general predictive-quality research establishes the value of ML and DL for mapping process conditions to quality outcomes. Pharmaceutical-specific reviews extend this work by incorporating PAT, QbD, and manufacturing unit operations. More recent studies increasingly emphasize hybrid models, digital twins, uncertainty, explainability, and regulatory implementation.

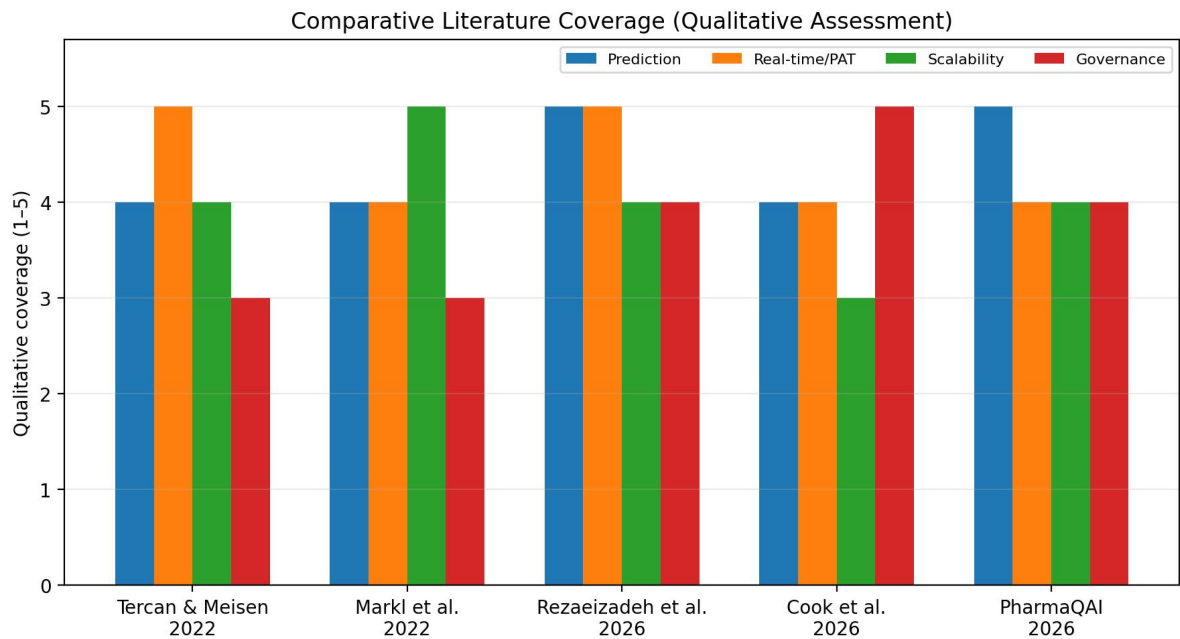


Figure 4. Qualitative comparison of selected literature across four platform-relevant dimensions. Scores are synthesis ratings by the present literature survey, not original experimental results.

8.1 Methodology Comparison

Neural networks are suitable for nonlinear relationships but can require significant data and may be difficult to interpret. Tree-based models can provide strong predictive performance and feature importance, particularly for structured process data. Deep learning and computer vision are appropriate for high-dimensional signals and images, while hybrid mechanistic–ML models may improve scientific consistency and extrapolation. Therefore, the proposed platform should support model selection based on the data modality and business objective rather than using one algorithm for every quality problem.

8.2 Data and Tool Comparison

The literature uses process parameters, PAT measurements, historical batch records, sensor streams, manufacturing datasets, and image data. This heterogeneity creates a major integration challenge. A

smart QA platform requires data ingestion, synchronization, cleaning, contextualization, and lineage tracking before model training. The data layer must also retain timestamps, batch identifiers, instrument information, and audit metadata to support traceability.

8.3 Performance and Scalability

Reported performance varies because studies use different datasets, targets, validation methods, and metrics. Consequently, direct comparison of numerical accuracy across papers is often inappropriate. More meaningful evaluation includes prediction error, sensitivity to deviations, false alarm rate, robustness across batches, latency, calibration, uncertainty, and performance under distribution shift. Scalability should also consider whether a model can be transferred across equipment, products, manufacturing sites, or process scales.

8.4 Governance and Compliance

The strongest recurring implementation theme is that AI performance must be combined with governance. A high-accuracy model may still be unsuitable if its input data cannot be trusted, its decisions cannot be explained, or its lifecycle cannot be validated and monitored. The proposed platform therefore includes access control, audit trails, version management, data lineage, model monitoring, human approval, and documented change management.

9. LIMITATIONS AND CHALLENGES IDENTIFIED

- Data silos: manufacturing, laboratory, maintenance, and quality systems may not share a unified data model.
- Limited labeled failure data: rare critical deviations may be too infrequent for conventional supervised learning.
- Data quality and integrity: missing values, sensor drift, inconsistent timestamps, and manual-entry errors can reduce reliability.
- Interpretability: black-box predictions may be difficult for quality professionals to justify.
- Uncertainty: a prediction should indicate confidence, especially near the edge of the training distribution.
- Model drift: changes in materials, equipment, suppliers, or process conditions can reduce performance over time.
- Transferability: a model trained for one product or site may not generalize to another.

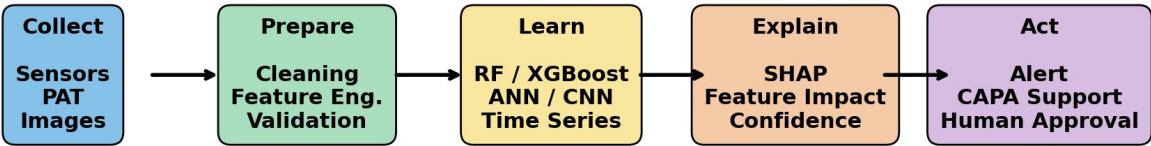
- Validation and lifecycle management: AI models require controlled development, testing, approval, monitoring, and change procedures.
- Cybersecurity and privacy: interconnected manufacturing systems increase the attack surface.
- Human factors: users require training and should understand AI limitations.

10. IDENTIFIED RESEARCH GAP

The literature provides strong evidence for individual technologies such as predictive ML, ANN-based PAT, computer vision, digital twins, and process modeling. However, a major gap remains in combining these capabilities into a single end-to-end quality assurance framework that is simultaneously predictive, explainable, uncertainty-aware, traceable, and aligned with regulated deployment. Many studies optimize model performance for a specific dataset or unit operation, whereas real manufacturing requires integration across the full data lifecycle.

The proposed research gap can therefore be stated as follows: there is insufficient evidence for a unified AI-based pharmaceutical manufacturing quality assurance platform that integrates multi-source manufacturing data, CQA prediction, real-time anomaly detection, explainable AI, risk prioritization, human-in-the-loop approval, model health monitoring, and audit-ready governance within one scalable architecture.

AI/ML Quality Intelligence Pipeline



Closed-loop improvement: verified outcomes are fed back for monitored model retraining and validation

Figure 5. Proposed AI quality intelligence pipeline from data collection to human-supervised action.

11. PROPOSED AI-BASED SMART PHARMACEUTICAL MANUFACTURING QUALITY ASSURANCE PLATFORM

11.1 Data Acquisition Layer

The platform receives data from PAT instruments, IoT sensors, SCADA or process control systems, MES, LIMS, equipment logs, environmental monitoring, batch records, and visual inspection systems.

Each incoming record should be associated with time, batch, product, process stage, source system, and quality context.

11.2 Data Management and Integration Layer

This layer performs data validation, cleaning, synchronization, missing-value handling, feature engineering, and secure storage. Data lineage is maintained so that every model prediction can be traced back to the source records and transformation steps used to create it.

11.3 AI and Analytics Layer

Different analytical services are used according to the quality problem. Regression predicts continuous CQAs such as assay or dissolution. Classification estimates pass/fail or deviation risk. Anomaly detection identifies unusual multivariate patterns. Time-series models monitor drift. Computer vision inspects packaging or surface defects. Optimization and hybrid models can recommend process settings within approved operating ranges.

11.4 Explainability and Uncertainty Layer

Feature-importance and explainability methods can show which variables contributed to a prediction. Confidence intervals, prediction intervals, or calibrated uncertainty estimates can help distinguish reliable predictions from cases requiring manual review. Out-of-distribution detection should flag situations that differ substantially from model training data.

11.5 Quality Decision and Human Oversight Layer

The platform converts technical predictions into quality information such as risk level, deviation alert, predicted CQA, confidence, and recommended investigation priority. Final decisions remain subject to authorized human review according to the applicable quality procedure. AI recommendations should not bypass required approval and release responsibilities.

11.6 Governance and Compliance Layer

This layer manages user access, audit trails, electronic records, model versions, data versions, validation evidence, monitoring thresholds, retraining approvals, and change history. The aim is to support trustworthy AI operation rather than simply maximize algorithmic accuracy.

12. RISK-BASED QUALITY MANAGEMENT

A risk-based approach can combine the likelihood and impact of a predicted event. For example, a low-probability, high-impact contamination risk may still require immediate escalation. The platform can prioritize investigations by combining process deviation probability, predicted CQA impact, equipment criticality, and confidence in the prediction.

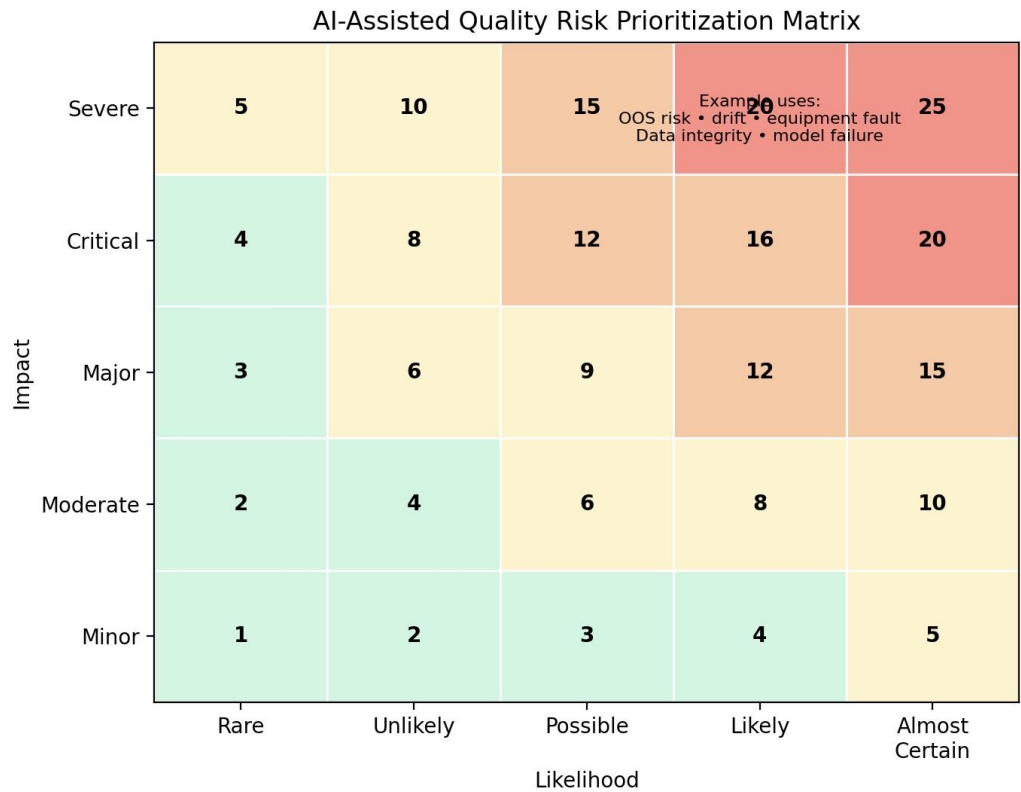


Figure 6. Example qualitative risk prioritization matrix for AI-assisted quality alerts.

13. FUNCTIONAL REQUIREMENTS

1. Acquire real-time and historical data from authorized manufacturing and quality sources.
2. Validate data completeness and flag missing, inconsistent, or abnormal records.
3. Predict selected critical quality attributes from process and material information.
4. Detect abnormal process patterns and early signs of quality drift.
5. Support image-based inspection where suitable visual data is available.
6. Calculate a quality risk score and prioritize alerts.
7. Provide explanations for model predictions and identify influential variables.
8. Display model confidence and flag low-confidence predictions.

9. Maintain audit trails for data, predictions, model versions, and user actions.
10. Support role-based access and authorized human approval.
11. Monitor model performance and detect degradation after deployment.
12. Generate dashboards and reports for quality teams and management.

14. NON-FUNCTIONAL REQUIREMENTS

- Reliability: stable operation and graceful handling of unavailable data sources.
- Scalability: support additional products, equipment, sensors, and models.
- Security: access control, authentication, encryption, and monitoring.
- Traceability: every important prediction traceable to its data and model version.
- Usability: dashboards that present quality information clearly.
- Maintainability: modular models and integrations for controlled updates.
- Performance: real-time alerts should meet process latency requirements.

15. PROPOSED METHODOLOGY

The following methodology is proposed for implementing the platform as a research prototype or industrial pilot.

1. Define the manufacturing use case, product scope, CQAs, CPPs, risk criteria, and intended users.
2. Collect and integrate historical batch, process, PAT, laboratory, equipment, and inspection data.
3. Perform data quality checks, remove duplicates, align timestamps, handle missing values, and document data lineage.
4. Conduct exploratory analysis to understand distributions, correlations, process stages, and known deviation patterns.
5. Create engineered features based on process knowledge and QbD understanding.
6. Split data using time-aware or batch-aware validation where appropriate to avoid information leakage.
7. Train baseline and advanced models for each defined task.
8. Evaluate models using task-appropriate metrics such as MAE/RMSE/R² for regression and precision/recall/F1/AUC for classification.
9. Assess robustness, calibration, uncertainty, false alarms, and performance on unseen batches.
10. Apply explainability methods to verify that model behavior is scientifically plausible.
11. Deploy approved models behind controlled APIs or services connected to a dashboard.
12. Monitor data drift, model performance, user feedback, and alert outcomes under a documented lifecycle process.

Proposed Implementation and Future Research Roadmap

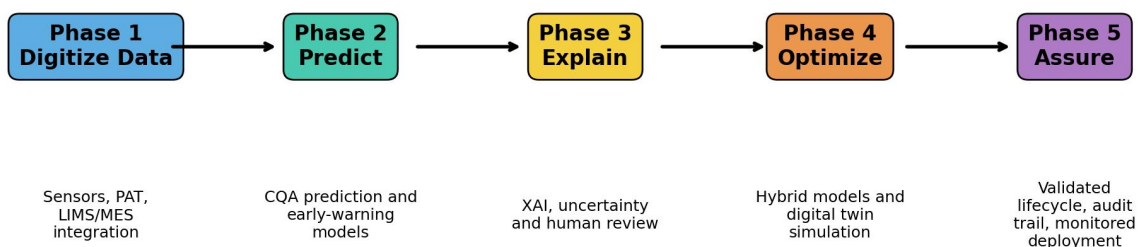


Figure 7. Proposed implementation and future research roadmap.

16. EVALUATION METRICS

16.1 Regression Quality Prediction

For continuous CQA prediction, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2) can be used. However, the platform should also evaluate whether prediction error is acceptable relative to the quality specification and process decision context.

16.2 Classification and Deviation Detection

Accuracy alone may be misleading when deviations are rare. Precision, recall, F1-score, ROC-AUC, precision-recall curves, sensitivity, specificity, and false alarm rate should be considered. High recall may be important for detecting serious deviations, while excessive false alarms can reduce user trust.

16.3 Operational Metrics

Operational evaluation can include alert latency, percentage of early detections, reduction in manual review effort, investigation prioritization quality, model availability, data completeness, and time required to identify probable root causes. These metrics connect technical performance to practical quality benefits.

17. EXPECTED BENEFITS

- Earlier detection of process drift and potential quality failures.
- Prediction of selected CQAs before final laboratory results are available.
- Reduced dependence on purely retrospective deviation investigation.
- Improved visibility of complex multivariate process relationships.
- Faster prioritization of high-risk batches and investigations.
- Better use of PAT, sensor, image, and historical manufacturing data.
- Improved decision support through explainable AI and confidence information.
- Potential reduction of waste, rework, unnecessary testing, and process variability.
- Support for continuous improvement and knowledge management.

18. FUTURE DIRECTIONS

Future work should focus on hybrid physics-informed and machine-learning models, uncertainty-aware prediction, federated or privacy-preserving learning across sites, validated digital twins, synthetic data for rare deviations, multimodal learning across numerical and image data, causal approaches for root-cause analysis, and standardized model lifecycle management. Research should also investigate how quality professionals interact with AI recommendations, because human factors, trust, and organizational processes are central to safe deployment.

A particularly important future direction is the development of benchmark datasets and evaluation protocols that reflect realistic pharmaceutical conditions. Models should be tested not only on randomly split datasets but also across time periods, equipment states, material changes, and distribution shifts. This will provide stronger evidence of reliability under real manufacturing variability.

19. CONCLUSION

The literature survey shows that AI and ML have significant potential to transform pharmaceutical quality assurance from a largely reactive activity into a more predictive, data-driven, and proactive process. Research supports the use of AI for quality prediction, PAT analytics, anomaly detection, process modeling, computer vision, optimization, and digital manufacturing. At the same time, the literature consistently identifies barriers involving data quality, limited datasets, interpretability, uncertainty, scalability, transferability, cybersecurity, validation, and governance.

The key research gap is not simply the need for a more accurate model. Instead, there is a need for an integrated platform that connects data acquisition, predictive analytics, anomaly detection, explainability, uncertainty, risk prioritization, human oversight, and compliant lifecycle management. The proposed AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform addresses this gap conceptually through a layered architecture and a risk-based, human-in-the-loop workflow. Such a system can provide meaningful decision support while preserving the traceability and accountability required in pharmaceutical quality operations.

Therefore, the proposed platform represents a software engineering approach to combining emerging AI technologies with responsible quality management. The next stage of research should implement a prototype using a suitable manufacturing dataset and evaluate the system through predictive performance, robustness, usability, explainability, operational benefit, and controlled model lifecycle criteria.

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